



UNIVERSITY EXAMINATIONS: 2014/2015
ORDINARY EXAMINATION FOR THE BACHELOR OF BUSINESS
INFORMATION TECHNOLOGY

BBIT 105 PROGRAMMING METHODOLOGY- DISTANCE
LEARNING

DATE: DECEMBER, 2014

TIME: 2 HOURS

INSTRUCTIONS: Answer Question ONE and any other TWO

QUESTION ONE [30 MARKS]

- a) State the values of the control variable x to be printed by each of the following statements in c.
 - i) for(x=2;x<=13;x+=2); [2 Marks]
 - ii) for (x=12;x>=2;x-=3) [2 Marks]
- b) A recursive function can be defined as a function which either calls itself or is in a potential cycle of function calls. Write a program in c that uses a recursive function to calculate a factorial of a number. [6 Marks]
- c) Design a program using pseudo code to calculate and display sum of even numbers between 2 and 30. [4 Marks]
- d) Write a program in c to implement the design in [c]. [4 Marks]
- e) Differentiate between IF and SWITCH selection structures in a c program. [4 Marks]
- f) A variable is an identifier that denotes a storage location used to store a data value. Outline three important rules for naming variables in c. [4 Marks]
- g) Describe imperative programming, and in particular highlight its distinctive features. [4 Marks]

QUESTION TWO [20 MARKS]

- a) Design a program using arrays to calculate and display the total weight in kilograms of forty six (46) passengers. [6 Marks]
- b) Write a program in c to implement the design in [a]. [8 Marks]
- c) The following statements were used to declare arrays. State the amount of memory allocated for each at compile time.

[3 Marks]

[3 Marks]

a) Discuss the term indirection as used in pointers. [4 Marks]

[4 Marks]

[2 Marks]

[2 Marks]

[2 Marks]

[2 Marks]

[8 Marks]

a) Define a string, hence state and explain the uses of any four functions used in string manipulations. [9 Marks]

[9 Marks]

```
#include<stdio.h>
#include<string.h>
int main()
{
    char strg1[100];
    char strg2[100];
    printf("\nEnter the first string : ");
    gets(strg1);
    printf("\nEnter the second string : ");
    gets(strg2);
    strcpy(strg2, strg1);
    strcpy(strg1, strg2);
    printf("\nThe value of first string is : %s\n", strg1);
    printf("\nThe value of the second string : %s\n", strg2);
    return (0);
}
```

}

[6 Marks]

- c) Write a program to calculate the amount **a** at the end of **t** years if the principle value **p** earns a compound interest at the rate **r**, using library function **math.h**. The amount is calculated using the formula $a = p(1+r)^n$. [5 Marks]

QUESTION FIVE [20 MARKS]

- (a) Explain the difference between fatal and non fatal error in a program. Give an example in each case. [6 marks]
(b) Identify and explain the errors in the following code snippets.

i) `int x = 5;
while(x > 0);

x--;`

[2 Marks]

ii) `int x = 5;
if (x = 6)
printf("x equals 6\n");`

[2 Marks]

- (c) Explain the differences and similarities between DO loop and WHILE loop. Present the general syntax for each structure. [6 marks]
(d) Explain the output of the following code in c.

```
#include <stdio.h>
int main()
{
    int i = 13;
    while(i>=1)
    {
        int k = i % 2;
        printf("%d\n", k);
        i--;
    }
    return 0;
}
```

[4 Marks]

